



EXPERIMENT NO-3

Study and construction of Half Adder using Basic Gates.



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Experiment No – 3, Study and construct a half adder using basic digital logic gates.

Theory:

A half-adder serves as an essential arithmetic component designed for the addition of two binary digits. This circuit incorporates two input ports, representing the two binary bits to be summed, and delivers a pair of outputs: one for the resulting SUM and the other for the CARRY generated during the addition process.

Figure 3.1 shows the block diagram of a half adder and figure 3.2 shows the truth table of a half-adder, showing all possible input combinations and the corresponding outputs.

The Boolean expressions for the SUM and CARRY outputs are given by the equations,

Let us denote the Sum as (S) and Carry as (C).

$$S = A'.B + A.B' \dots\dots\dots (3.1)$$

$$C = A.B \dots\dots\dots (3.2)$$

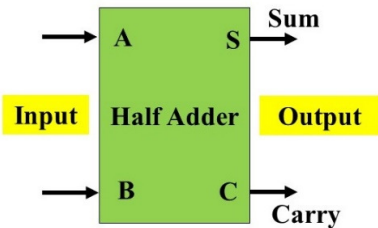


Figure (3.1), block diagram of half adder.

A	B	Sum (S)	Carry (C)
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Figure (3.2), Truth Table of half adder.

As evident from the truth table the **Sum (S)** can be implemented using a XOR logical gate and **Carry (C)** can be implemented using two inputs AND logic gate. The figure (3.3) shows the logical circuits using XOR & AND logic gates.

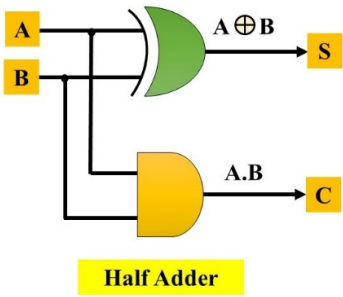


Figure (3.3), Logical circuit diagram using XOR & AND logic gate.

The circuit in figure (3.3), can be redrawn using only basic logic gates as shown in the figure (3.4).

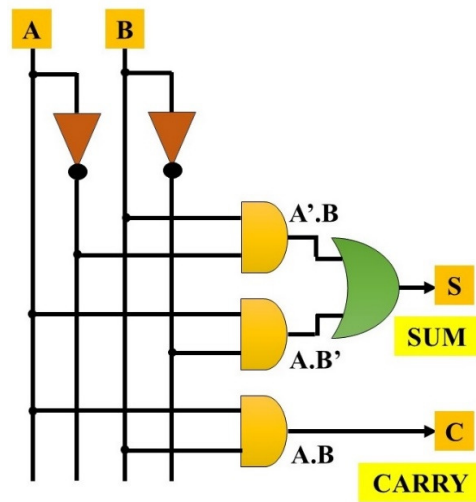


Figure (3.4), Half adder using basic logic gates.

Implementation

The circuit can be implemented using a Hex inverter (7404), Quad 2-input AND gate (7408) and Quad 2-input OR gate (7432).

The circuit given in the figure (3.5) shows the actual implementation of half adder.

Step 1: Two inverters are used to complement the inputs.

Step 2: One of the AND gate is connected in such a fashion to get the output as $A'.B$ and the other is used to generate $A.B'$.

Step 3: The outputs of the AND gates are Ored to get the final SUM (S) output.

Step 4: third AND gate is used to generate $A.B$ which is the CARRY (C) out.

Semiconductors and Instruments required

1. DC +5V regulated power supply.
2. Wish/Bread board.
3. Jumper wires.
4. Digital voltmeter.
5. Quad 2-input OR gate (7432)
6. Quad 2-input AND gate (7408)
7. Hex inverter (7404)

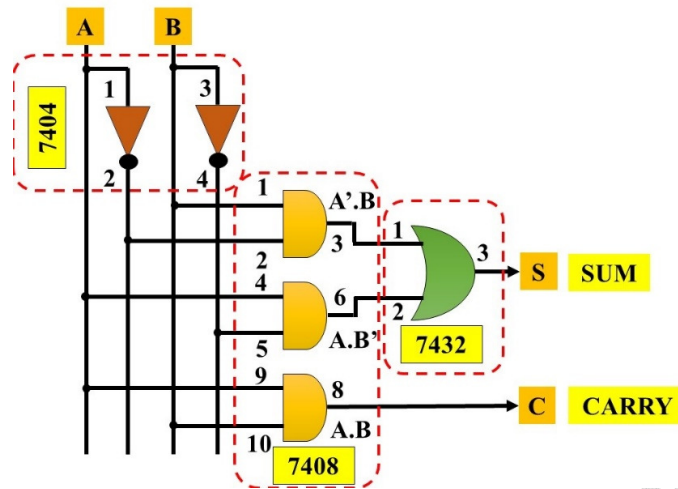


Figure (3.5), Half adder circuit using basic logic gates.

Result/Observation

Truth table for the half adder using basic logic gates.

Logical Input		Logical output		Output voltage	
A	B	SUM (S)	CARRY (C)	SUM (S)	CARRY (C)
0	0	0	0		
0	1	1	0		
1	0	1	0		
1	1	0	1		

Discussion

1. The power supply used must be regulated to +5V dc.
2. The jumpers used on the bread board must be short.
3. All the IC used must be properly seated on the bread board along with proper connection with regulated power supply.
4. The inputs to be connected to logic - 0 must not be left floating. For logic-1 and logic-0 we need to connect the inputs either to high voltage i.e. **+5V (V_{CC})** or to the ground potential.
5. Outputs can be visualized by connecting LED. Those must be connected via current limiting resistor.